AMRITA SCHOOL OF ENGINEERING, BENGALURU

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B. TECH. IV SEMESTER (EAC) AY 2025-2026

Electronics System Design and Automation

PORTFOLIO ON COLOUR SORTING SYSTEM

JANUARY TO APRIL 2026

Submitted by

BL.EN.U4EAC24018 Ch.Sriya Bharathi
BL.EN.U4EAC24028 G.Prasanna
BL.EN.U4EAC24044 M. Padmaja
BL.EN.U4EAC24046 N.Abhinay
BL.EN.U4EAC24057 R.Bindu
BL.EN.U4EAC24058 Rohit M

TITLE: GEMS SORTING SYSTEM USING ARDUINO NANO AND TCS3200

PROJECT OVERVIEW:

This project implements an Automated colour sorting system using Arduino nano and TCS3200 colour sensor . The system detects object colour using frequency based sensing and sorts the gems into their respective bins using servo motors.

PROBLEM STATEMENT:

Manual sorting is time consuming and error prone . This project rectifies it and sorts using embedded systems.

OBJECTIVE:

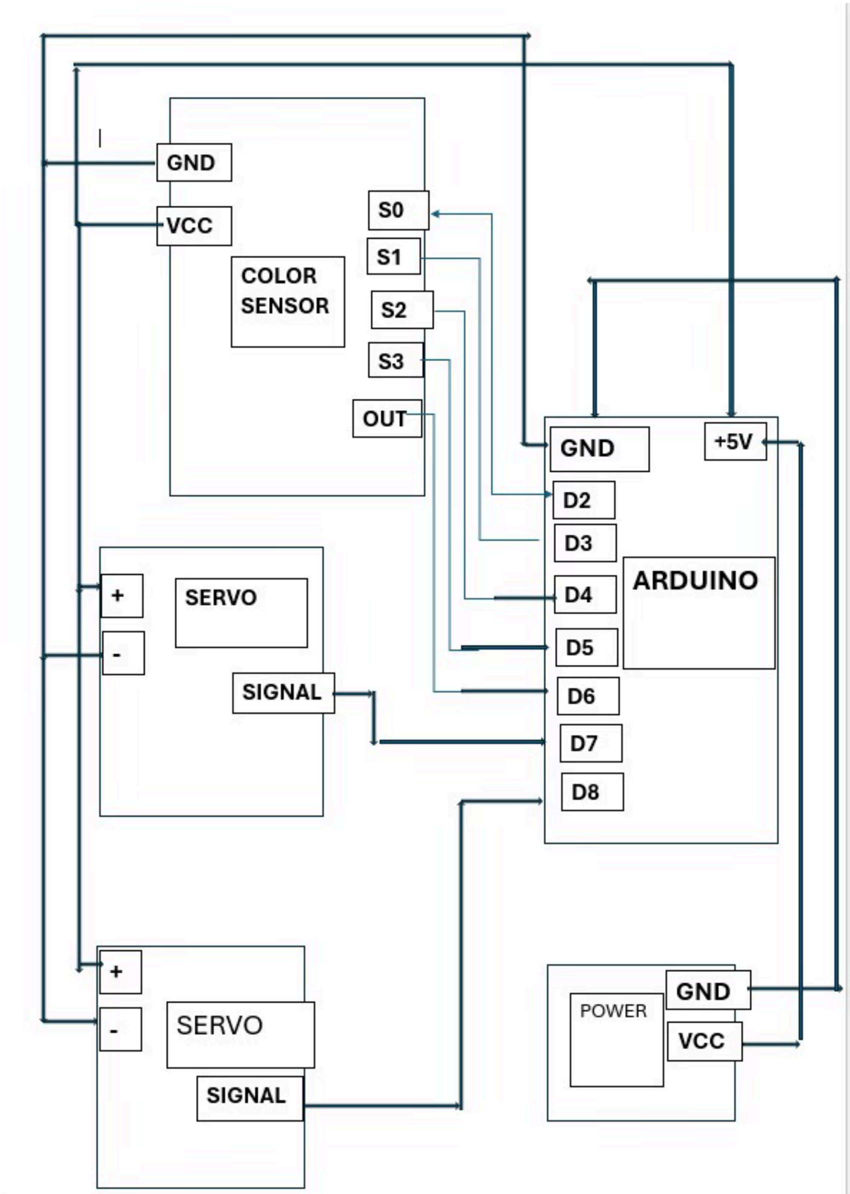
- Detect colour using colour sensor.
- Process RGB values using Arduino nano(i.e converting colour into frequency)
- Sorts object automatically
- Improve accuracy using calibration.

COMPONENTS USED:

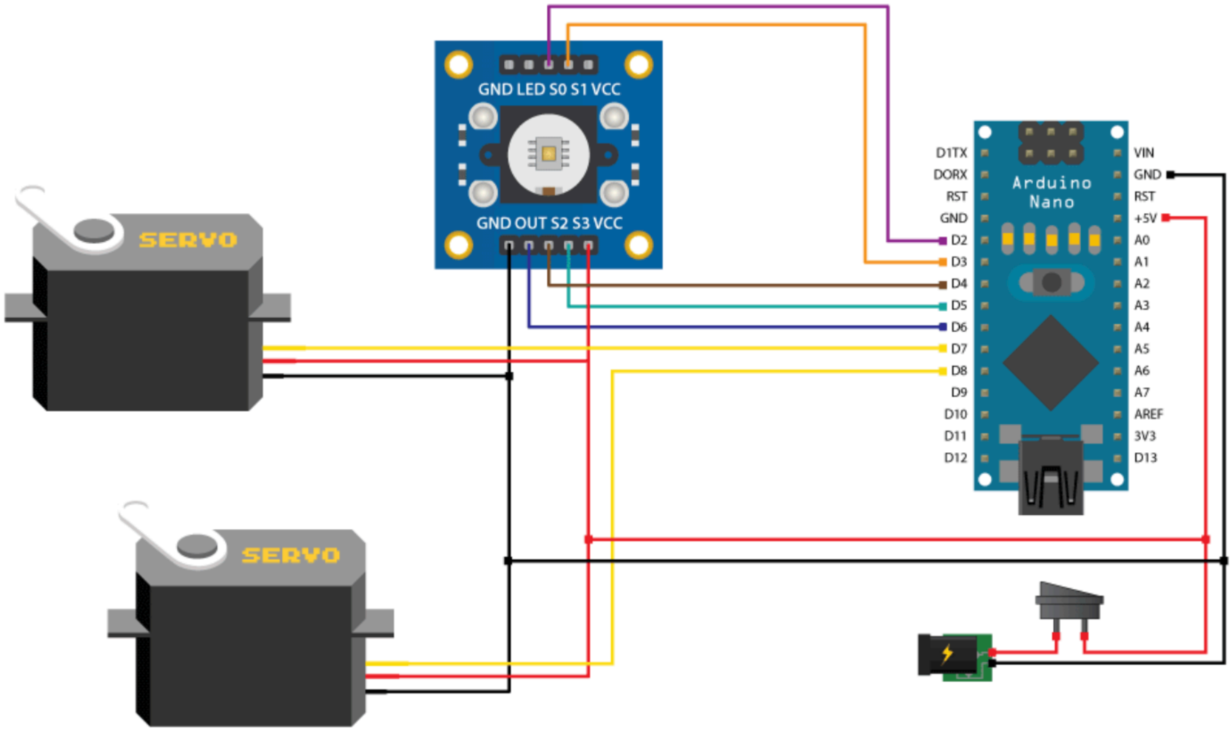
COMPONENT	PURPOSE	COST
Arduino Nano	Control unit	195/-
TCS3200 Sensor	Colour detection	320/-
Servo Motors(2)	Sensing+Sorting	120/- x 2
Breadboard	Circuit connections	60/-
Jumper wires.	Wiring	74/-
Lead wire	Soldering	20/-
DC adapter	5V Supply	130/-
Arduino Cable		43/-
Power Jack		20/-
Shoe Box	Layout	20/-

TOTAL COST: 1,102/-

BLOCK DIAGRAM:


GND
VCC
S0
S1
COLOR
SENSOR
S2
S3
OUT
GND
+5V
D2
D3
ARDUINO
D4
+
SERVO
-
D5
SIGNAL
D6
D7
D8
+
-
SERVO
SIGNAL
GND
POWER
VCC

CIRCUIT DIAGRAM:



WORKING PRINCIPLE:

The system works on the principle of colour detection using frequency conversion and mechanical sorting using servo motors.

1. Color Sensing (TCS3200)

- The TCS3200 colour sensor contains an array of photodiodes with Red, Green, Blue filters.
- When a gem is placed in front of the sensor:
  - It is illuminated by white light (from onboard LEDs).
  - The reflected light from the gem falls on the photodiodes.
- Based on the intensity of reflected RGB light, the sensor generates a square wave output frequency.

Higher intensity → Higher frequency

Lower intensity → Lower frequency

So basically:

- Colour → Light intensity → Frequency signal

2. Frequency to Digital Conversion (Arduino Nano)

- The output frequency from the sensor is given to the Arduino Nano.
- The Arduino:
  - Measures frequency for Red, Green, Blue channels one by one.
  - Converts these frequencies into RGB values.
  - Compares them with pre-calibrated threshold values.

3. Decision Making

- Based on the detected colour Arduino decides , which bin the gem belongs to.

This is basically a classification step.

4. Sorting Mechanism (Servo Motors)

There are totally 2 servos used ;

Servo 1 (Feeding Mechanism):
- Picks or releases one gem at a time to ensure controlled input.

Servo 2 (Sorting Mechanism):
- Rotates to a specific angle depending on the detected colour.
- Servo works using PWM (Pulse Width Modulation) signals from Arduino.

5. Continuous Operation

- After sorting one gem , next gem is processed.
- This creates an automated sorting loop.

ALGORITHM:

Steps are as follows:
- Start
- Initialise sensor and servo
- Place object
- Read Red frequency
- Read Green frequency
- Read Blue frequency
- Compare values
- Arduino identifies colour
- Rotate Servo(slider)
- Drop the object
- Repeat the process

CODE:

```
#include <Servo.h>

#define S0 2
#define S1 3
#define S2 4
#define S3 5
#define sensorOut 6

Servo topServo;
Servo bottomServo;

int frequency = 0;
int color=0;

void setup() {
```

```
  pinMode(S0, OUTPUT);
  pinMode(S1, OUTPUT);
  pinMode(S2, OUTPUT);
  pinMode(S3, OUTPUT);
  pinMode(sensorOut, INPUT);

  digitalWrite(S0, HIGH);
  digitalWrite(S1, LOW);

  topServo.attach(7);
  bottomServo.attach(8);

  Serial.begin(9600);
}

void loop() {

  topServo.write(115);
  delay(1000);

  for(int i = 115; i > 65; i--) {
    topServo.write(i);
    delay(2);
  }
  delay(1000);

  color = readColor();
  delay(10);

  switch (color) {
    case 1:
    bottomServo.write(75);
    break;

    case 2:
    bottomServo.write(100);
    break;

    case 3:
    bottomServo.write(125);
    break;

    case 4:
    bottomServo.write(150);
    break;

    case 5:
    bottomServo.write(175);
```

```
    break;

    case 6:
    bottomServo.write(0);
    break;

    case 0:
    break;
  }
  delay(1000);

  for(int i = 65; i > 29; i--) {
    topServo.write(i);
    delay(2);
  }
  delay(200);

  for(int i = 29; i < 115; i++) {
    topServo.write(i);
    delay(2);
  }
  color=0;
}

int readColor() {
  digitalWrite(S2, LOW);
  digitalWrite(S3, LOW);
  frequency = pulseIn(sensorOut, LOW);
  int R = frequency;
  Serial.print("R= ");
  Serial.print(frequency);
  Serial.print("  ");
  delay(50);

  digitalWrite(S2, HIGH);
  digitalWrite(S3, HIGH);

  frequency = pulseIn(sensorOut, LOW);
  int G = frequency;

  Serial.print("G= ");
  Serial.print(frequency);
  Serial.print("  ");
  delay(50);

  digitalWrite(S2, LOW);
 digitalWrite(S3, HIGH);
```

```
  frequency = pulseIn(sensorOut, LOW);
  int B = frequency;

  Serial.print("B= ");
  Serial.print(frequency);
  Serial.println("  ");
  delay(50);

  if(R<60 && R>32 && G<65 && G>40){
    color = 1;
  }
  if(G<66 && G>43 && B<56 && B>35){
    color = 2;
  }
  if(R<54 && R>45 && G<64 && G>55){
    color = 3;
  }
  if(R<53 && R>44 && G<61 && G>45){
    color = 4;
  }
  if(R<53 && R>46 && G<65 && G>55){
    color = 5;
  }
  if (G<66 && G>57 && B<54 && B>45){
    color = 6;
  }
  return color;
}
```

ADVANTAGES:

- Faster sorting.
- Reduces human error.
- Low Cost.

LIMITATIONS:

- Affected by outside bright light.
- Difficult for dark colours.
- Calibration issues.